

Type	Transform	Inverse transform
Fourier transform	$F(u) = \int_{-\infty}^{\infty} f(x) e^{iux} dx$	$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(u) \cdot \bar{e}^{iux} du$
Fourier cosine transform	$F_c(u) = \int_0^{\infty} f(x) \cdot \cos(ux) dx$	$f(x) = \frac{2}{\pi} \int_0^{\infty} F_c(u) \cos(ux) du$
Fourier sine transform	$F_s(u) = \int_0^{\infty} f(x) \sin(ux) dx$	$f(x) = \frac{2}{\pi} \int_0^{\infty} F_s(u) \sin(ux) du$

$$\begin{aligned}
 1. \quad i &= \sqrt{-1} \\
 i^2 &= -1 \\
 i^3 &= i \cdot i^2 \\
 &= i(-1) \\
 i^3 &= -i
 \end{aligned}$$

$$i^4 = 1$$

$$3. \frac{i}{i^3} = \frac{1}{-i} = i$$

$$4. e^{in\theta} = \cos(n\theta) + i\sin(n\theta)$$

$$5. e^{-in\theta} = \cos(n\theta) - i\sin(n\theta)$$

$$2. \frac{1}{i} = \frac{1}{i} \times \frac{i}{i} = \frac{i}{i^2} = -i$$

$$6. e^{in\theta} + e^{-in\theta} = 2\cos(n\theta)$$

$$7. e^{in\theta} - e^{-in\theta} = 2i\sin(n\theta)$$

$$8. \int e^{ax} \cos(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a\cos(bx) + b\sin(bx))$$

$$9. \int e^{ax} \sin(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a\sin(bx) - b\cos(bx))$$

$$10. e^{i\theta} = \cos\theta + i\sin\theta$$

$$11. e^{-i\theta} = \cos\theta - i\sin\theta$$

01. Find the Fourier Transformation of $f(x) = e^{-|x|}$

$\Rightarrow$ Given :-

$$f(x) = e^{-|x|} = \begin{cases} e^{-x} & \text{for } x > 0 \\ e^{+x} & \text{for } x < 0 \end{cases}$$

$$= \begin{cases} e^{-x} & \text{for } x > 0 \\ e^{+x} & \text{for } x < 0 \end{cases}$$

$$F(u) = \int_{-\infty}^{\infty} f(x) \cdot e^{iux} dx$$

$$= \int_{-\infty}^0 f(x) e^{iux} dx + \int_0^{\infty} f(x) e^{iux} dx$$

$$= \int_{-\infty}^0 e^{+x} e^{iux} dx + \int_0^{\infty} e^{-x} e^{iux} dx$$

$$= \int_{-\infty}^0 e^{x+iux} dx + \int_0^{\infty} e^{-x+iux} dx$$

$$= \int_{-\infty}^0 e^{x(1+iu)} dx + \int_0^{\infty} e^{-(1-iu)x} dx$$

$$= \int_{-\infty}^0 e^{(1+iu)x} dx + \int_0^{\infty} e^{-(1-iu)x} dx$$

$$= \left[\frac{e^{(1+iu)x}}{1+iu} \right]_{-\infty}^0 + \left[\frac{e^{-(1-iu)x}}{-(1-iu)} \right]_0^{\infty}$$

$$= \frac{1}{1+iu} (e^0 - e^{-\infty}) - \frac{1}{1-iu} (e^{-\infty} - e^0)$$

$$= \frac{1}{1+iu} (1-0) - \frac{1}{(1-iu)} (0-1)$$

$$= \frac{1}{1+iu} + \frac{1}{1-iu} \quad \because e^{\pm\infty} = \frac{1}{e^{\mp\infty}} = 0$$

$$= \frac{(1-iu) + (1+iu)}{1^2 - (iu)^2} = \frac{1-iu+1+iu}{1+u^2}$$

$$\boxed{F(u) = \frac{2}{1+u^2}} \quad \left[e^{ax} = \frac{1}{a} \right]$$

IMP

02. Find the Fourier transform of $f(x) = \begin{cases} 1 & \text{for } |x| \leq a \\ 0 & \text{for } |x| > a \end{cases}$ Hence

$$\text{evaluate } \int_0^{\infty} \frac{\sin x}{x} dx$$

$\Rightarrow$ Given

$$f(x) = \begin{cases} 1 & \text{for } -a \leq x \leq a \\ 0 & \text{otherwise} \end{cases}$$

we have

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{iux} dx$$

$$= \int_{-\infty}^{-a} f(x) e^{iux} dx + \int_{-a}^a f(x) e^{iux} dx + \int_a^{\infty} f(x) e^{iux} dx$$

$$= \int_{-a}^a e^{iux} dx$$

$$= \int_{-a}^a 1 \cdot e^{iux} dx$$

$$= \int_{-a}^a e^{iux} dx$$

$$= \left[\frac{e^{iux}}{iu} \right]_{-a}^a$$

$$= \frac{1}{iu} [e^{iua} - e^{-iua}]$$

$$= \frac{1}{iu} [e^{iua} - e^{-iua}] \quad [\because e^{iua} - e^{-iua} = 2i \sin(ua)]$$

$$\boxed{F(u) = \frac{2 \sin(au)}{u}}$$

$$\text{To evaluate } \int_0^{\infty} \frac{\sin x}{x} dx$$

the inverse fourier
transform

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(u) \cdot e^{-iux} du$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2 \sin(au)}{u} e^{-iux} du$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2 \sin(au)}{u} e^{-iux} du$$

put $x=0$ we have $f(x)=1$

$$- a \leq x \leq a$$

$$f(0)=1$$

$$f(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2 \sin(au)}{u} \cdot e^0 du$$

$$1 = \frac{1}{\pi} \int_{-\infty}^{\infty} \left(\frac{\sin(au)}{u} \right) du$$

$$1 = \frac{2}{\pi} \int_0^{\infty} \frac{\sin(au)}{u} du$$

$$\frac{\pi}{2} = \int_0^{\infty} \frac{\sin(au)}{u} du$$

put $a=1$

$$\int_0^{\infty} \frac{\sin(u)}{u} du = \frac{\pi}{2}$$

replace u by x

$$\boxed{\int_0^{\infty} \frac{\sin(x)}{x} dx = \frac{\pi}{2}}$$

Reference

To check the eq. even or odd

$$f(u) = \frac{\sin(au)}{u}$$

$$f(-u) = \frac{\sin(-au)}{-u}$$

$$= \frac{-\sin(au)}{-u}$$

$$= \frac{\sin(au)}{u}$$

$$= f(u)$$

Here $f(u)$ is even

eg. Find the Fourier Transform of

$$f(x) = x e^{-|x|}$$

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} x e^{+x} & x < 0 \\ x e^{-x} & x > 0 \end{cases}$$

$$F(u) = \int_{-\infty}^{\infty} f(x) \cdot e^{+iux} dx$$

$$= \int_{-\infty}^0 f(x) e^{iux} dx + \int_0^{\infty} f(x) e^{iux} dx$$

$$= \int_{-\infty}^0 x e^{x+iux} dx + \int_0^{\infty} x e^{-x+iux} dx$$

$$= \int_{-\infty}^0 x \cdot e^{(1+iu)x} dx + \int_0^{\infty} x \cdot e^{-(1-iu)x} dx$$

$$= \left[x \cdot \frac{e^{(1+iu)x}}{1+iu} - (1) \cdot \frac{e^{(1+iu)x}}{(1+iu)^2} \right]_{-\infty}^0 + \left[x \cdot \frac{e^{-(1-iu)x}}{-(1-iu)} - (1) \frac{e^{-(1-iu)x}}{(1-iu)^2} \right]_0^{\infty}$$

$$= \frac{-1}{(1+iu)^2} \left(e^{(1+iu)x} \right)_{-\infty}^0 - \frac{1}{(1-iu)^2} \left(e^{-(1-iu)x} \right)_0^{\infty}$$

$$= \frac{-1}{(1+iu)^2} (e^0 - e^{-\infty}) - \frac{1}{(1-iu)^2} (e^{-\infty} - e^0)$$

$$\begin{aligned} (a+b)^2 - (a-b)^2 &= (a^2+b^2) - (a^2-b^2) \\ &= a^2+b^2+2ab - a^2+b^2-2ab \\ &= 4ab \end{aligned}$$

$$= \frac{-1}{(1+iu)^2} (1-0) - \frac{1}{(1-iu)^2} (0-1)$$

$$= \frac{-1}{(1+iu)^2} + \frac{1}{(1-iu)^2} \Rightarrow \frac{-(1-iu)^2 + (1+iu)^2}{(1+iu)^2 (1-iu)^2}$$

$$= \frac{4iu}{(1-(iu)^2)^2} = \frac{4iu}{(1+u^2)^2}$$

Note $\therefore \int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & \text{for } f(x) \text{ is even f}^n \\ 0 & \text{for } f(x) \text{ is odd f}^n \end{cases}$

Q4. Find the Fourier transform of $f(x) = \begin{cases} 1-|x| & \text{for } |x| \leq 1 \\ 0 & \text{for } |x| > 1 \end{cases}$ & hence deduce that $\int_0^\infty \frac{\sin^2 t}{t^2} dt = \pi/2$

$\Rightarrow$ Given

$$f(x) = \begin{cases} 1-|x| & \text{for } -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$F(u) = \int_{-\infty}^{\infty} f(x) \cdot e^{iux} dx$$

$$= \int_{-\infty}^{-1} f(x) e^{iux} dx + \int_{-1}^{+1} f(x) e^{iux} dx + \int_{+1}^{\infty} f(x) e^{iux} dx$$

$$= \int_{-1}^1 (1-|x|) e^{iux} dx$$

$$= \int_{-1}^0 (1-(-x)) e^{iux} dx + \int_0^1 (1-x) e^{iux} dx \quad \because |x| = \begin{cases} -x & x < 0 \\ x & x > 0 \end{cases}$$

$$= \int_{-1}^0 (1+x) \cdot e^{iux} dx + \int_0^1 (1-x) \cdot e^{iux} dx$$

$$= \left[(1+x) \frac{e^{iux}}{iu} - (1) \frac{e^{iux}}{(iu)^2} \right]_{-1}^0 + \left[(1-x) \frac{e^{iux}}{iu} - (-1) \frac{e^{iux}}{(iu)^2} \right]_0^1$$

$$= \frac{1}{iu} \left[(1+x) e^{iux} \right]_{-1}^0 + \frac{1}{u^2} \left[e^{iux} \right]_{-1}^0 + \frac{1}{iu} \left[(1-x) e^{iux} \right]_0^1 - \frac{1}{u^2} (e^{iux})_0^1$$

$$= \frac{1}{iu} (1e^0 - 0) + \frac{1}{u^2} (e^0 - e^{-iu}) + \frac{1}{iu} (0 - 1e^0) - \frac{1}{u^2} (e^{iu} - e^0)$$

$$= \frac{1}{iu} + \frac{1}{u^2} (1 - e^{-iu}) - \frac{1}{iu} - \frac{1}{u^2} (e^{iu} - 1)$$

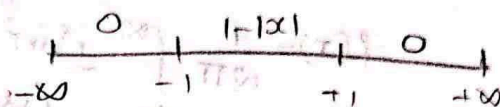
$$= \frac{1}{u^2} - \frac{e^{-iu}}{u^2} - \frac{e^{iu}}{u^2} + \frac{1}{u^2}$$

$$= \frac{2}{u^2} - \frac{1}{u^2} (e^{iu} + e^{-iu}) = \frac{2}{u^2} - \frac{1}{u^2} (2 \cos u)$$

$$= \frac{2}{u^2} (1 - \cos u)$$

$$= \frac{2}{u^2} [2 \sin^2(u/2)]$$

$$F(u) = \frac{4 \sin^2(u/2)}{u^2}$$



$$1 - \cos \theta = 2 \sin^2 \theta/2$$

$$1 - \cos u = 2 \sin^2(u/2)$$

we have inverse fourier transform

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(u) \cdot e^{-iux} du$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{4\sin^2(u/2)}{u^2} \cdot e^{-iux} du$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2(u/2)}{(u/2)^2} e^{-iux} du$$

put $x=0$ $f(x) = 1-|x|$ for $-1 \leq x \leq 1$

$$f(0) = 1-|0|$$

$$f(0) = 1$$

$$f(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2(u/2)}{(u/2)^2} e^0 du$$

$$1 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2(u/2)}{(u/2)^2} du$$

put $u/2 = t$

$$u = 2t \Rightarrow du = 2dt$$

$$1 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2(t)}{t^2} \cdot 2dt$$

$$1 = \frac{2}{2\pi} \int_{-\infty}^{\infty} \frac{\sin^2(t)}{t^2} dt$$

$$1 = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin^2(t)}{t^2} dt$$

$$1 = \frac{1}{\pi} \times 2 \int_0^{\infty} \frac{\sin^2(t)}{t^2} dt$$

$$\boxed{\int_0^{\infty} \frac{\sin^2(t)}{t^2} dt = \frac{\pi}{2}}$$

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Q5. Find the fourier transform of $f(x) = \begin{cases} 1-x^2 & \text{for } |x| < 1 \\ 0 & \text{for } |x| \geq 1 \end{cases}$
Hence evaluate

(i) $\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} dx$ (ii) $\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cdot \cos\left(\frac{x}{2}\right) dx$

$$f(x) = \begin{cases} 1-x^2 & -1 < x < 1 \\ 0 & \text{otherwise } x \leq -1 \text{ \& } x \geq 1 \end{cases}$$

$$-1 < x < 1$$

$$\text{otherwise } x \leq -1 \text{ \& } x \geq 1$$

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{iux} dx$$

$$|x| < 1, -1 < x < 1$$

$$|x| \geq 1, x \leq -1 \text{ \& } x \geq 1$$

$$0 \quad 1-x^2 \quad 0$$

$$= \int_{-\infty}^{-1} f(x) e^{iux} dx + \int_{-1}^1 f(x) e^{iux} dx + \int_1^{\infty} f(x) e^{iux} dx$$

$$= \int_{-1}^1 (1-x^2) e^{iux} dx$$

$$= \left[(1-x^2) \frac{e^{iux}}{iu} \right]_{-1}^1 - (0-2x) \left(\frac{e^{iux}}{i^2 u^2} \right) + (-2) \left(\frac{e^{iux}}{i^3 u^3} \right) \quad \begin{matrix} i^2 = -1 \\ i^3 = i^2 \cdot i \\ = -1 \cdot i = -i \end{matrix}$$

$$= \frac{1}{iu} \left[(1-x^2) e^{iux} \right]_{-1}^1 + \frac{2}{i^2 u^2} \left[x e^{iux} \right]_{-1}^1 - \frac{2}{i^3 u^3} \left[e^{iux} \right]_{-1}^1$$

$$= \frac{1}{iu} [0-0] + \frac{2}{(-1)u^2} [1e^{iu(1)} - (-1)e^{iu(-1)}] - \frac{2}{(-i)u^3} [e^{iu(1)} - e^{iu(-1)}]$$

$$= -\frac{2}{u^2} [e^{iu} + e^{-iu}] + \frac{2}{iu^3} [e^{iu} - e^{-iu}] \quad \begin{matrix} e^{ix} + e^{-ix} = 2\cos x \\ e^{ix} - e^{-ix} = 2i\sin x \end{matrix}$$

$$= -\frac{2}{u^2} [2\cos u] + \frac{2}{iu^3} [2i\sin u] = -\frac{4\cos u}{u^2} + \frac{4\sin u}{u^3}$$

$$F(u) = -\frac{4u\cos u}{u^3} + \frac{4\sin u}{u^3} = \frac{-4u\cos u + 4\sin u}{u^3}$$

$$F(u) = -4 \left[\frac{u\cos u - \sin u}{u^3} \right]$$

Inverse Fourier Transform is

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(u) e^{-iux} du$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\frac{u\cos u - \sin u}{u^3} \right) e^{-iux} du$$

$$f(x) = \frac{-2}{\pi} \int_{-\infty}^{\infty} \frac{u\cos u - \sin u}{u^3} e^{-iux} du \quad -2 \cdot 0$$

(i) $x=0$ in $f(x)$ (ii) eqⁿ (i)

$$f(0) = \frac{-2}{\pi} \int_{-\infty}^{\infty} \frac{u\cos u - \sin u}{u^3} e^{-iu(0)} du$$

$$f(0) = 1$$

$$\begin{matrix} x=0 & f(x) = 1-x^2 \\ f(0) = 1-0^2 \\ f(0) = 1 \end{matrix}$$

$$1 = \frac{-2}{\pi} \int_{-\infty}^{\infty} \frac{u \cos u - \sin u}{u^3} du$$

$\therefore \frac{u \cos u - \sin u}{u^3}$ is even

$$1 = \frac{-2}{\pi} (2) \int_0^{\infty} \frac{u \cos u - \sin u}{u^3} du$$

$$\int_0^{\infty} \frac{u \cos u - \sin u}{u^3} du = -\frac{\pi}{4}$$

replace u by x

$$\therefore \int_0^{\infty} \frac{x \cos x - \sin x}{x^3} dx = -\frac{\pi}{4}$$

(ii) let $x = 1/2$ in eqⁿ (i)

$$f(1/2) = \frac{-2}{\pi} \int_{-\infty}^{\infty} \frac{u \cos u - \sin u}{u^3} e^{-iu(1/2)} du$$

$$f(1/2) = 1 - (1/2)^2 = \frac{3}{4}$$

$$f(1/2) = 1 - x^2$$

$$= 1 - (1/2)^2 = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\frac{3}{4} = \frac{-2}{\pi} \int_{-\infty}^{\infty} \frac{u \cos u - \sin u}{u^3} e^{-i(u/2)} du$$

$$e^{-ix} = \cos x - i \sin x$$

$$\frac{3}{4} = \frac{-2}{\pi} \int_{-\infty}^{\infty} \frac{u \cos u - \sin u}{u^3} \left(\cos\left(\frac{u}{2}\right) - i \sin\left(\frac{u}{2}\right) \right) du$$

$$\frac{3}{4} = \frac{-2}{\pi} \int_{-\infty}^{\infty} \frac{u \cos u - \sin u}{u^3} \cos\left(\frac{u}{2}\right) du$$

$$\frac{3}{4} = \frac{-2}{\pi} (2) \int_0^{\infty} \frac{u \cos u - \sin u}{u^3} \cos\left(\frac{u}{2}\right) du$$

$\therefore \frac{u \cos u - \sin u}{u^3} \cos\left(\frac{u}{2}\right)$ is even

$$\int_0^{\infty} \frac{u \cos u - \sin u}{u^3} \cos\left(\frac{u}{2}\right) du = -\frac{3\pi}{16}$$

$\therefore$ replace u as x

$$\therefore \int_0^{\infty} \frac{x \cos x - \sin x}{x^3} \cos\left(\frac{x}{2}\right) dx = -\frac{3\pi}{16}$$

6. Find the complex Fourier transform of $f(x) = e^{-ax^2}$ where a is +ve constant. Hence deduce that $e^{-x^2/2}$ is self-orthogonal / reciprocal in respect of complex Fourier Transform

$$\Rightarrow \text{Given } f(x) = e^{-a^2 x^2}$$

we have fourier transform

$$F(u) = \int_{-\infty}^{\infty} f(x) e^{iux} dx$$

$$= \int_{-\infty}^{\infty} e^{-a^2 x^2} e^{iux} dx = \int_{-\infty}^{\infty} e^{(-a^2 x^2 + iux)} dx \quad \div \& \text{ multiply by 2} \\ \& \text{ by } \left(\frac{i u}{2a^2}\right)^2$$

$$= \int_{-\infty}^{\infty} e^{-a^2 \left[x^2 - 2x \left(\frac{i u}{2a^2}\right) + \left(\frac{i u}{2a^2}\right)^2 - \left(\frac{i u}{2a^2}\right)^2 \right]} dx \quad i^2 = -1$$

$$= \int_{-\infty}^{\infty} e^{-a^2 \left[\left(x - \frac{i u}{2a^2}\right)^2 + \left(\frac{u}{2a^2}\right)^2 \right]} dx$$

$$= \int_{-\infty}^{\infty} e^{-a^2 \left(x - \frac{i u}{2a^2}\right)^2} \cdot e^{-a^2 \left(\frac{u}{2a^2}\right)^2} dx$$

$$= \int_{-\infty}^{\infty} e^{-a^2 \left(x - \frac{i u}{2a^2}\right)^2} \cdot e^{-a^2 \left(\frac{u}{2a^2}\right)^2} dx \cdot e^{-\frac{u^2}{4a^2}} \int_{-\infty}^{\infty} e^{-a^2 \left(x - \frac{i u}{2a^2}\right)^2} dx$$

$$\text{put } a \left(x - \frac{i u}{2a^2}\right) = t$$

$$a dx = dt$$

$$dx = \frac{dt}{a}$$

$\therefore t$ is also varies from $-\infty$ to ∞

$$e^{-\frac{u^2}{4a^2}} \int_{-\infty}^{\infty} e^{-t^2} \frac{dt}{a}$$

$$F(u) = \frac{e^{-\frac{u^2}{4a^2}}}{a} \int_{-\infty}^{\infty} e^{-t^2} dt$$

$$F(u) = \frac{e^{-\frac{u^2}{4a^2}}}{a} \cdot \sqrt{\pi}$$

$$\therefore \int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\pi}$$

$$\text{we have } F[e^{-a^2 x^2}] = \frac{e^{-\frac{u^2}{4a^2}}}{a} = \sqrt{\pi}$$

$$\text{put } a^2 = 1/2$$

$$F[e^{-x^2/2}] = \frac{e^{-\frac{u^2}{4 \times 1/2}}}{(1/\sqrt{2})} \cdot \sqrt{\pi}$$

$$F[e^{-x^2/2}] = \sqrt{2\pi} \cdot e^{-\frac{u^2}{2}}$$

replace u by x

$$F\left[e^{-\frac{x^2}{2}}\right] = \sqrt{2\pi} e^{-\frac{x^2}{2}}$$

$\therefore$ Hence $e^{-\frac{x^2}{2}}$ is self reciprocal under complex Fourier transform.

V.V.1

7. Find the Fourier sine & cosine transform of

$$f(x) = e^{-ax} \quad a > 0. \text{ Hence deduce that } \int_0^{\infty} \frac{\cos nx}{a^2 + x^2} dx = \frac{\pi}{2a} e^{-an}$$

$$dx = \frac{\pi}{2a} e^{-an}$$

$$\Rightarrow \text{Given } f(x) = e^{-ax}$$

$$F_s(u) = \int_0^{\infty} f(x) \cdot \sin(ux) dx$$

$$F_c(u) = \int_0^{\infty} f(x) \cos(ux) dx$$

$$F_s(u) = \int_0^{\infty} e^{-ax} \cdot \sin(ux) dx$$

$$\left[\frac{e^{-ax}}{(-a^2) + u^2} [-a \sin(ux) - u \cos(ux)] \right]_0^{\infty} \quad \begin{matrix} a = a \\ b = u \end{matrix} \frac{u}{a^2 + u^2}$$

$$\frac{1}{a^2 + u^2} \left[e^{-ax} (-a \sin(ux) - u \cos(ux)) \right]_0^{\infty}$$

$$F_s(u) = \frac{1}{a^2 + u^2} \left[e^{-\infty} - e^0 (-a \sin(0) - u \cos(0)) \right]$$

$$F_s(u) = \frac{u}{a^2 + u^2}$$

$$f(x) = e^{-ax}$$

$$F_c(u) = \int_0^{\infty} f(x) \cdot \cos(ux) dx$$

$$= \int_0^{\infty} e^{-ax} \cdot \cos(ux) dx$$

$$= \left[\frac{e^{-ax}}{(-a^2) + u^2} (-a \cos(ux) + u \sin(ux)) \right]_0^{\infty}$$

$$= \frac{1}{a^2 + u^2} \left[e^{-\infty} - e^0 (-a \cos(0) + u \sin(0)) \right]$$

$$F_c(u) = \frac{a}{a^2 + u^2}$$

which inverse Fourier cosine transform,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F(u) \cdot \cos(ux) du$$

$$e^{-ax} = \frac{2}{\pi} \int_0^{\infty} \frac{a}{a^2 + u^2} \cos(ux) du$$

$$\frac{\pi}{2a} e^{-ax} = \int_0^{\infty} \frac{\cos(ux)}{a^2 + u^2} du$$

$$\int_0^{\infty} \frac{\cos(ux)}{a^2 + u^2} du = \frac{\pi}{2a} e^{-ax}$$

replace u by x & x by m

$$\boxed{\int_0^{\infty} \frac{\cos(mx)}{a^2 + x^2} dx = \frac{\pi}{2a} e^{-am}}$$

8. Find the fourier sine and cosine transform of
 $2e^{-3x} + 3e^{-2x}$

$\Rightarrow$ Given $f(x) = 2e^{-3x} + 3e^{-2x}$

we have fourier sine & cosine transform

$$F_s[u] = \int_0^{\infty} f(x) \sin(ux) dx \quad \text{--- (1)} \quad F_c[u] = \int_0^{\infty} f(x) \cos(ux) dx \quad \text{--- (2)}$$

$$\begin{aligned} \textcircled{1} \Rightarrow F_s[u] &= \int_0^{\infty} (2e^{-3x} + 3e^{-2x}) \cdot \sin(ux) dx \\ &= 2 \int_0^{\infty} e^{-3x} \sin(ux) dx + 3 \int_0^{\infty} e^{-2x} \sin(ux) dx \end{aligned}$$

$$= 2 \left[\frac{e^{-3x}}{(-3)^2 + u^2} (-3 \sin ux - u \cos ux) \right]_0^{\infty} +$$

$$3 \left[\frac{e^{-2x}}{(-2)^2 + u^2} (-2 \sin ux - u \cos ux) \right]_0^{\infty}$$

$$\therefore \int_0^{\infty} e^{ax} \sin(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$$

$$= \frac{2}{9 + u^2} \left[e^{-\infty} (-3 \sin 0 - u \cos 0) - e^0 (-3 \sin 0 - u \cos 0) \right] + \frac{3}{4 + u^2} \left[e^{-\infty} (-2 \sin 0 - u \cos 0) - e^0 (-2 \sin 0 - u \cos 0) \right]$$

$$= \frac{2u}{9 + u^2} + \frac{3u}{4 + u^2}$$

$$\boxed{F_s[u] = u \left[\frac{2}{9 + u^2} + \frac{3}{4 + u^2} \right]}$$

$$\textcircled{2} \Rightarrow F_c[u] = \int_0^{\infty} f(x) \cos(ux) dx.$$

$$= \int_0^{\infty} (2e^{-3x} + 3e^{-2x}) \cos(ux) dx$$

$$= 2 \int_0^{\infty} e^{-3x} \cos(ux) dx + 3 \int_0^{\infty} e^{-2x} \cos(ux) dx$$

$$e^{ax} \cos(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$$

$$= 2 \left[\frac{e^{-3x}}{(-3)^2 + u^2} (-3 \cos ux + u \sin ux) \right]_0^{\infty} + 3 \left[\frac{e^{-2x}}{(-2)^2 + u^2} (-2 \cos ux + u \sin ux) \right]_0^{\infty}$$

$$= \frac{2}{9+u^2} [e^{-\infty} - e^0 (-3 \cos 0 + u \sin 0)] + \frac{3}{4+u^2} [e^{-\infty} - e^0 (-2 \cos 0 + u \sin 0)]$$

$$= \frac{2 \times 3}{9+u^2} + \frac{3 \times 2}{4+u^2} = \frac{6}{9+u^2} + \frac{6}{4+u^2}$$

$$\boxed{F_c[u] = 6 \left[\frac{1}{9+u^2} + \frac{1}{4+u^2} \right]}$$

Imp

9. Find the fourier sine transform of the function

$$f(x) = e^{-|x|} \text{ hence evaluate } \int_0^{\infty} \frac{x \sin(mx)}{1+x^2} dx, m > 0$$

$$\Rightarrow \text{Given } f(x) = e^{-|x|}$$

$$F_s(u) = \int_0^{\infty} f(x) \sin(ux) dx. \quad |x| = \begin{cases} -x & x < 0 \\ x & x \geq 0 \end{cases}$$

$$= \int_0^{\infty} e^{-|x|} \sin(ux) dx$$

$$= \int_0^{\infty} e^{-x} \sin(ux) dx$$

$$e^{ax} \sin(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$$

$$= \left[\frac{e^{-x}}{(-1)^2 + u^2} (-\sin ux - u \cos ux) \right]_0^{\infty} = \frac{1}{1+u^2} [e^{-\infty} - e^0 (-\sin 0 - u \cos 0)]$$

$$= \frac{1}{1+u^2} (-1(-u)) \Rightarrow \boxed{F_s(u) = \frac{u}{1+u^2}}$$

Inverse fourier sine transform

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F_s[u] \cdot \sin(ux) du.$$

$$e^{-|x|} = \frac{2}{\pi} \int_0^{\infty} \frac{u}{1+u^2} \cdot \sin(ux) du.$$

$$\frac{\pi}{2} e^{-|x|} = \int_0^{\infty} \frac{u \sin(ux)}{(1+u^2)} du$$

put $x=m$ $u \rightarrow$

$$\int_0^{\infty} \frac{u \sin(mu)}{(1+u^2)} du = \frac{\pi}{2} \cdot e^{-m}$$

change u to x .

$$\boxed{\int_0^{\infty} \frac{x \sin(mx)}{(1+x^2)} dx = \frac{\pi}{2} \cdot e^{-m}}$$

Imp.

10. Find the Fourier cosine transform of

$$f(x) = \begin{cases} x & \text{for } 0 < x < 1 \\ 2-x & \text{for } 1 < x < 2 \\ 0 & \text{for } x > 2 \end{cases}$$

$\Rightarrow$ Given.

$$F_c[u] = \int_0^{\infty} f(x) \cdot \cos(ux) dx$$

$$= \int_0^1 x \cos(ux) dx + \int_1^2 (2-x) \cos(ux) dx + \int_2^{\infty} 0 \cos(ux) dx$$

$$= \int_0^1 x \cos(ux) dx + \int_1^2 (2-x) \cos(ux) dx$$

$$= \left[x \frac{\sin(ux)}{u} - (1) \left(-\frac{\cos(ux)}{u^2} \right) \right]_0^1 + \left[(2-x) \frac{\sin(ux)}{u} - (-1) \left(-\frac{\cos(ux)}{u^2} \right) \right]_1^2$$

$$= \frac{1}{u} (x \sin ux)_0^1 + \frac{1}{u^2} (\cos ux)_0^1 + \frac{1}{u} [(2-x) \sin(ux)]_1^2 - \frac{1}{u^2} (\cos ux)_1^2$$

$$= \frac{1}{u} (\sin u - 0) + \frac{1}{u^2} (\cos u - \cos 0) + \frac{1}{u} (0 - \sin u) - \frac{1}{u^2} (\cos 2u - \cos u)$$

$$= \frac{\sin u}{u} + \frac{1}{u^2} (\cos u - 1) - \frac{\sin u}{u} - \frac{1}{u^2} \cos 2u + \frac{1}{u^2} \cos u$$

$$= \frac{1}{u^2} \cos u - \frac{1}{u^2} - \frac{1}{u^2} \cos 2u + \frac{1}{u^2} \cos u$$

$$= \frac{2}{u^2} \cos u - \frac{1}{u^2} (1 + \cos 2u) = \frac{2 \cos(u)}{u^2} - \frac{1}{u^2} (2 \cos^2 u)$$

$$= \frac{2}{u^2} (1 - \cos^2 u) = \frac{2}{u^2} \cos u (1 - \cos u) = \frac{2}{u^2} \cos u 2 \sin^2(u/2)$$

$$\boxed{F_c[u] = \frac{4 \cos u \sin^2(u/2)}{u^2}}$$

Zawamio

Imp 11. Solve integral equation $\int_0^{\infty} f(x) \cos(\alpha x) dx = \begin{cases} 1-\alpha & \text{for } 0 \leq \alpha \leq 1 \\ 0 & \text{for } \alpha > 1 \end{cases}$

hence deduce the value of $\int_0^{\infty} \frac{\sin^2(t)}{t^2} dt$

$\Rightarrow$ Given $\int_0^{\infty} f(x) \cos(\alpha x) dx = \begin{cases} 1-\alpha & \text{for } 0 \leq \alpha \leq 1 \\ 0 & \text{for } \alpha > 1 \end{cases}$

$$F_c[\alpha] = \begin{cases} 1-\alpha & \text{for } 0 \leq \alpha \leq 1 \\ 0 & \text{for } \alpha > 1 \end{cases}$$

we have

$$\begin{aligned} f(x) &= \frac{2}{\pi} \int_0^{\infty} F_c[\alpha] \cos(\alpha x) d\alpha \\ &= \frac{2}{\pi} \left[\int_0^1 (1-\alpha) \cos(\alpha x) d\alpha + \int_1^{\infty} 0 \cos(\alpha x) d\alpha \right] \\ &= \frac{2}{\pi} \left[(1-\alpha) \frac{\sin(\alpha x)}{\alpha} - (-1) \left(-\frac{\cos(\alpha x)}{x^2} \right) \right]_0^1 \\ &= \frac{2}{\pi x} \left[(1-\alpha) \sin(\alpha x) \right]_0^1 - \frac{2}{\pi x^2} \left[\cos(\alpha x) \right]_0^1 \\ &= \frac{2}{\pi x} (0 \sin x - 0) - \frac{2}{\pi x^2} (\cos x - \cos 0) \\ &= -\frac{2}{\pi x^2} (\cos x - 1) = \frac{2}{\pi x^2} (1 - \cos x) \end{aligned}$$

$$f(x) = \frac{2}{\pi x^2} (2 \sin^2 x/2) = \frac{4 \sin^2(x/2)}{\pi x^2} //$$

$$F_c[\alpha] = \int_0^{\infty} f(x) \cos(\alpha x) dx$$

$$= \int_0^{\infty} \frac{4 \sin^2(x/2)}{\pi x^2} \cos(\alpha x) dx$$

$$\because 1 - \cos x = 2 \sin^2 x/2$$

$$\pi F_c[\alpha] = \int_0^{\infty} \frac{\sin^2(x/2)}{(x/2)^2} \cos(\alpha x) dx$$

$$\therefore \text{put } \frac{x}{2} = t \quad \begin{matrix} \text{when} \\ x=0, t=0 \\ x=\infty, t=\infty \end{matrix}$$

$$dx = 2dt$$

$$\pi F_c[\alpha] = \int_0^{\infty} \frac{\sin^2(t)}{t^2} \cos(2\alpha t) \cdot 2 dt$$

put $\alpha = 0$, $F_c[\alpha] = 1 - \alpha$ for $0 \leq \alpha \leq 1$, $F_c[0] = 1 - 0 = 1$

$$\pi(1) = 2 \int_0^{\infty} \frac{\sin^2 t}{t^2} \cos(0) dt$$

$$\boxed{\int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \pi/2}$$

12. Find the Fourier sine transform of $\frac{e^{-ax}}{x}$, $a > 0$

$$\Rightarrow \text{Given } f(x) = \frac{e^{-ax}}{x}$$

$$F_s[u] = \int_0^{\infty} f(x) \sin(ux) dx$$

$$F_s[u] = \int_0^{\infty} \frac{e^{-ax}}{x} \cdot \sin(ux) dx$$

D.w.r.t 'u'

$$\frac{d}{du} (F_s[u]) = \int_0^{\infty} \frac{e^{-ax}}{x} \cdot \frac{d}{du} (\sin(ux)) dx$$

$$= \int_0^{\infty} \frac{e^{-ax}}{x} x (\cos ux) \cdot dx$$

$$= \int_0^{\infty} e^{-ax} \cos ux dx$$

$$= \frac{e^{-ax}}{(a^2 + u^2)} \left[(-a \cos ux + u \sin ux) \right]_0^{\infty} \quad \begin{matrix} a = -a \\ b = u \end{matrix}$$

$$= \frac{1}{a^2 + u^2} \left[\cancel{e^{-\infty}}^0 (-a \cos \infty + u \sin \infty) - e^0 (-a \cos 0 + u \sin 0) \right]$$

$$= \frac{1}{a^2 + u^2} (-1(-a))$$

$$\frac{d}{du} [F_s[u]] = \frac{a}{a^2 + u^2}$$

on integrating w.r.t 'u'

$$F_s[u] = \int \frac{a}{a^2 + u^2} du$$

$$F_s[u] = \tan^{-1} \left(\frac{u}{a} \right) + C \quad \text{--- (1)}$$

put $u = 0$

$$F_s[0] = \tan^{-1}(0) + C$$

$$0 = 0 + C$$

$$\boxed{C = 0}$$

$$\textcircled{1} \Rightarrow \boxed{F_s[u] = \tan^{-1} \left(\frac{u}{a} \right)}$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right)$$